

FUNCTIONAL INEQUALITIES AND SYNCHRONIZATION FOR THE MEASURE-VALUED SOLUTIONS OF STOCHASTIC DIFFERENTIAL EQUATIONS WITH INTERACTION

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ABSTRACT. We study the measure-valued solution to the stochastic differential equations with interaction. Under regularity assumptions, we obtain moment estimates for the maximal Jacobian norm on compact sets, which imply the functional inequalities. We also establish an averaged Jacobian-energy bound under one-sided derivative assumptions. Finally, for a kernel-type interacting model satisfying a pairwise dissipativity condition, we prove exponential decay of the expected pairwise variance, yielding convergence to the center of mass.

CONTENTS

1.	Introduction	1
2.	Pathwise Jacobian Bounds and Functional Inequality Transfer	4
2.1.	The Jacobian flow	4
2.2.	Transfer of functional inequalities	5
2.3.	Proof of the compact-set Jacobian estimate	7
3.	The averaged Jacobian	11
4.	Internal fluctuations	12
4.1.	Dissipativity assumptions	12
4.2.	Proof	13
	References	16

1. INTRODUCTION

We consider stochastic differential equations with interaction that were proposed by Andrey A. Dorogovtsev in [Dorogovtsev, 2003],

$$(1) \quad \begin{cases} dX_t(u) &= a(X_t(u), \mu_t) dt + \sum_{\ell=1}^r \sigma_\ell(X_t(u)) dB_t^\ell, \\ X_0(u) &= u, \\ \mu_t &= \mu_0 \circ X_t^{-1}. \end{cases}$$

Here $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$ is a deterministic initial probability measure, where $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of probability measures on $\mathbb{R}^d$ with finite second moment. $a : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}^d$ is the drift and the diffusion coefficient $\sigma_\ell : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\ell = 1, \dots, r$, is state-dependent and non-constant. Let $(\Omega, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions and carrying a r -dimensional Brownian motion $B = (B^1, \dots, B^r)$.

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Assumption 1 (Well-posedness). *There exists $L < \infty$ such that, for all $x, y \in \mathbb{R}^d$ and $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$|a(x, \mu) - a(y, \nu)| + \left(\sum_{\ell=1}^r |\sigma_\ell(x) - \sigma_\ell(y)|^2 \right)^{1/2} \leq L(|x - y| + W_2(\mu, \nu))$$

Under Assumption 1, the SDEWI (1) admits a unique global solution; see [Dorogovtsev, 2023, Theorem 2.3.1].

Unlike a classical stochastic differential equation, the drift depends on the pushforward of the entire system through the random measure μ_t . This dependence represents the interaction between particles. For each realization $\omega \in \Omega$, the map

$$T_t(\omega, \cdot) : u \mapsto X_t(u, \omega)$$

acts on the initial measure by pushforward,

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1},$$

so the evolution is best viewed as a random transport of measures by a stochastic flow (1). Whenever the flow is differentiable with respect to the initial label, we write

$$J_t(u) := \partial_u T_t(u),$$

and consider the three quantities described below.

In Section 2, we study $\mathcal{K}_t(K)$ on the compact set K . Firstly, we consider the maximal Jacobian norm $\mathcal{K}_t(K)$

$$\mathcal{K}_t(K) := \sup_{u \in K} \|J_t(u)\|_{\text{op}}.$$

We prove moment estimates for this random constant on compact sets and show that it governs the pathwise transfer of logarithmic Sobolev, concentration, and transport inequalities from μ_0 to μ_t .

Definition 1.1 (Logarithmic Sobolev inequality). Denoting by $\mathcal{C}_c^1(\mathbb{R}^d)$ the set of continuously differentiable compactly supported functions from $\mathbb{R}^d$ to $\mathbb{R}$, for $g \geq 0$ with $\int g d\mu = 1$ we define the entropy

$$\text{Ent}_\mu(g) := \int g \log g d\mu.$$

A probability measure μ on $\mathbb{R}^d$ is said to satisfy a logarithmic Sobolev inequality with constant $C > 0$ if for every $h \in \mathcal{C}_c^1(\mathbb{R}^d)$ with $\int h^2 d\mu = 1$ one has

$$(2) \quad \text{Ent}_\mu(h^2) \leq C \int_{\mathbb{R}^d} |\partial h|^2 d\mu.$$

Definition 1.2 (Talagrand transport inequality). Let $p \geq 1$. A probability measure μ on $\mathbb{R}^d$ is said to satisfy the Talagrand transport inequality $T_p(C_T)$ with constant $C_T > 0$ if for every probability measure ν on $\mathbb{R}^d$ with finite p -th moment,

$$(3) \quad W_p(\nu, \mu)^2 \leq 2C_T \text{Ent}(\nu \mid \mu),$$

where

$$\text{Ent}(\nu \mid \mu) = \begin{cases} \int_{\mathbb{R}^d} \log \left(\frac{d\nu}{d\mu} \right) d\nu, & \nu \ll \mu, \\ +\infty, & \text{otherwise.} \end{cases}$$

In Section 3, we replace the supremum over labels by the averaged Jacobian energy $\mathcal{E}_t$. The Jacobian energy $\mathcal{E}_t$

$$\mathcal{E}_t := \int_{\mathbb{R}^d} \|J_t(u)\|_{\text{HS}}^2 \mu_0(du),$$

over the initial mass μ_0 . Averaged Jacobian energy can be estimated under weaker one-sided derivative conditions.

Finally, in Section 4, we exploit the specific kernel-type interacting model, we impose a pairwise dissipativity condition and prove exponential decay of $\mathbb{E}\mathcal{D}_t$. This decay means that the transported labels approach one another in mean square and that the random measure synchronizes around its random center of mass.

The pairwise energy $\mathcal{D}_t$

$$\mathcal{D}_t := \iint_{\mathbb{R}^d \times \mathbb{R}^d} |T_t(u) - T_t(v)|^2 \mu_0(du) \mu_0(dv).$$

measures deformation at finite scale by averaging the squared distance between two transported labels. Pairwise energy can be controlled directly through dissipativity of the dynamics, without obtaining a uniform Jacobian estimate

2. PATHWISE JACOBIAN BOUNDS AND FUNCTIONAL INEQUALITY TRANSFER

Firstly, we establish $\mathcal{K}_t(K)$, then use it to transfer functional inequalities from the initial measure μ_0 to the random pushforward measure μ_t . The technical proof of the compact-set Jacobian estimate is postponed to the final subsection.

Assumption 2 (Spatial differentiability). *For every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, the map $x \mapsto a(x, \mu)$ belongs to $\mathcal{C}^1(\mathbb{R}^d; \mathbb{R}^d)$, and*

$$\sigma_\ell \in \mathcal{C}^1(\mathbb{R}^d; \mathbb{R}^d), \quad \ell = 1, \dots, r.$$

By Assumptions 1 and 2,

$$(4) \quad \|\partial_x a(x, \mu)\|_{\text{op}} \leq L_x, \quad \sum_{\ell=1}^r |\partial_x \sigma_\ell(x) z|^2 \leq L_\sigma^2 |z|^2$$

for all $x, z \in \mathbb{R}^d$ and $\mu \in \mathcal{P}_2(\mathbb{R}^d)$

2.1. The Jacobian flow. We use the Jacobian SDE from [Dorogovtsev and Weiß, 2023]. For equation (1), it has the following form.

Theorem 2.1 (Jacobian equation). *Assume Assumptions 1 and 2. Then the solution admits a modification such that, for almost every ω and every $t \geq 0$ its Jacobian $J_t(u)$ satisfies*

$$(5) \quad \begin{cases} dJ_t(u) = A_t(u) J_t(u) dt + \sum_{\ell=1}^r C_{\ell,t}(u) J_t(u) dB_t^\ell, \\ J_0(u) = I, \end{cases}$$

where

$$A_t(u) := \partial_x a(X_t(u), \mu_t), \quad C_{\ell,t}(u) := \partial_x \sigma_\ell(X_t(u)).$$

Definition 2.1 (Random Jacobian constant). For a nonempty set $K \subset \mathbb{R}^d$, define

$$\mathcal{K}_t(K) := \sup_{u \in K} \|J_t(u)\|_{\text{op}}.$$

If K is compact, continuity of $u \mapsto J_t(u)$ implies $\mathcal{K}_t(K) < \infty$. This compact-set quantity is used because a corresponding supremum over all of $\mathbb{R}^d$ need not be finite. The counterexamples of Mohammed–Scheutzow show that even for globally Lipschitz coefficients one may fail to have a finite global constant $\sup_{u \in \mathbb{R}^d} \|J_t(u, \omega)\|_{\text{op}}$ for a fixed (t, ω) .

Theorem 2.2 (Moments of the compact-set maximal Jacobian norm). *Assume Assumptions 1 and 2. Suppose in addition that there exists a constant $H < \infty$ such that, for all $x, y, z \in \mathbb{R}^d$ and $\mu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$(6) \quad |(\partial_x a(x, \mu) - \partial_x a(y, \mu))z| + \left(\sum_{\ell=1}^r |(\partial_x \sigma_\ell(x) - \partial_x \sigma_\ell(y))z|^2 \right)^{1/2} \leq H|x - y||z|.$$

Then, for every nonempty compact $K \subset \mathbb{R}^d$ and every $p \geq 1$, there exist constants $C_{p,K}, c_p < \infty$ such that

$$(7) \quad \mathbb{E}[\mathcal{K}_t(K)^p] \leq C_{p,K} e^{c_p t}, \quad t \geq 0.$$

The proof is given in Section 2.3.

2.2. Transfer of functional inequalities. Throughout this subsection, assume that $\text{supp}(\mu_0)$ is compact and set

$$(8) \quad K_0 := \text{co}(\text{supp}(\mu_0)), \quad \kappa_t(\omega) := \mathcal{K}_t(K_0).$$

Since K_0 is compact and convex, for every fixed $t \geq 0$ and almost every ω ,

$$(9) \quad \text{Lip}(T_t(\omega, \cdot)|_{K_0}) \leq \kappa_t(\omega).$$

Indeed, for $u, v \in K_0$,

$$\begin{aligned} |T_t(\omega, u) - T_t(\omega, v)| &\leq \int_0^1 \|J_t(v + s(u - v), \omega)\|_{\text{op}} |u - v| ds \\ &\leq \kappa_t(\omega) |u - v|, \end{aligned}$$

because $v + s(u - v) \in K_0$ for every $s \in [0, 1]$.

Corollary 2.1 (Pathwise logarithmic Sobolev inequality). *Suppose that μ_0 satisfies $\text{LSI}(C_0)$. Then, for every fixed $t \geq 0$ and almost every ω ,*

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega, \cdot)^{-1}$$

satisfies

$$\text{LSI}(C_0 \kappa_t(\omega)^2).$$

Proof. Fix $t \geq 0$ and ω such that $u \mapsto T_t(\omega, u)$ is of class $\mathcal{C}^1$, and write

$$T := T_t(\omega, \cdot), \quad J(u) := J_t(u, \omega), \quad \kappa := \kappa_t(\omega).$$

Let $g \in \mathcal{C}_c^1(\mathbb{R}^d)$ satisfy

$$\int_{\mathbb{R}^d} g^2 d\mu_t(\omega) = 1.$$

Choose $\chi \in \mathcal{C}_c^1(\mathbb{R}^d)$ such that $\chi = 1$ on a neighborhood of K_0 , and set

$$h(u) := \chi(u)g(T(u)).$$

Since μ_0 is supported on K_0 ,

$$h(u) = g(T(u)), \quad \partial h(u) = J(u)^\top \partial g(T(u)), \quad u \in K_0.$$

Consequently,

$$\begin{aligned} \text{Ent}_{\mu_t(\omega)}(g^2) &= \text{Ent}_{\mu_0}(h^2) \\ &\leq C_0 \int_{K_0} |J(u)^\top \partial g(T(u))|^2 \mu_0(du) \\ &\leq C_0 \kappa^2 \int_{K_0} |\partial g(T(u))|^2 \mu_0(du) \\ &= C_0 \kappa^2 \int_{\mathbb{R}^d} |\partial g(x)|^2 \mu_t(\omega, dx). \end{aligned}$$

□

Proposition 2.1 (Pathwise concentration transfer). *Suppose that μ_0 satisfies*

$$(10) \quad \mu_0(f - \mu_0(f) \geq r) \leq \Psi(r), \quad r \geq 0,$$

for every 1-Lipschitz function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, where $\Psi : [0, \infty) \rightarrow [0, 1]$ is nonincreasing.

Then, for every fixed $t \geq 0$, almost every ω , every 1-Lipschitz function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, and every $r > 0$,

$$(11) \quad \mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) \leq \Psi\left(\frac{r}{\kappa_t(\omega)}\right),$$

whenever $\kappa_t(\omega) > 0$.

If $\kappa_t(\omega) = 0$, then $\mu_t(\omega)$ is a Dirac mass and the probability on the left-hand side is zero.

Proof. Fix $t \geq 0$ and ω for which (9) holds, and write

$$T := T_t(\omega, \cdot), \quad \kappa := \kappa_t(\omega).$$

If $\kappa = 0$, then (9) shows that T is constant on K_0 . Since μ_0 is supported on K_0 , $\mu_t(\omega)$ is a Dirac mass.

Suppose that $\kappa > 0$, let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be 1-Lipschitz, and define on K_0

$$h(u) := \frac{g(T(u))}{\kappa}.$$

By (9), h is 1-Lipschitz on K_0 . By the McShane extension theorem, there exists a 1-Lipschitz function $\tilde{h} : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $\tilde{h} = h$ on K_0 .

Since $\mu_0(K_0) = 1$,

$$\mu_0(\tilde{h}) = \mu_0(h) = \frac{1}{\kappa} \mu_t(\omega)(g).$$

Therefore,

$$\begin{aligned} \mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) \\ &= \mu_0\left(\tilde{h} - \mu_0(\tilde{h}) \geq \frac{r}{\kappa}\right) \\ &\leq \Psi\left(\frac{r}{\kappa}\right). \end{aligned}$$

□

Remark 2.1 (Concentration as a consequence of LSI). If μ_0 satisfies a logarithmic Sobolev inequality, then it satisfies a Gaussian concentration inequality. Therefore, Corollary 2.1 yields a pathwise Gaussian concentration inequality for $\mu_t(\omega)$, with random concentration constant controlled by

$$\kappa_t(\omega)^2 = \mathcal{K}_t(K_0)^2.$$

We next record the analogous stability property for Talagrand transport inequalities.

Proposition 2.2 (Transport inequality under a localized Lipschitz pushforward). *Let μ be a probability measure satisfying $T_p(C_T)$ for some $p \geq 1$, and suppose that $\text{supp}(\mu) \subset K$ for some compact set $K \subset \mathbb{R}^d$.*

Let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a Borel map whose restriction to K is Lipschitz, and set

$$L_K(T) := \sup_{\substack{u, v \in K \\ u \neq v}} \frac{|T(u) - T(v)|}{|u - v|}.$$

Then the pushforward measure $\eta := \mu \circ T^{-1}$ satisfies

$$T_p(C_T L_K(T)^2).$$

Proof. Let λ be a probability measure on $\mathbb{R}^d$. If $\lambda \ll \eta$, then $\text{Ent}(\lambda \mid \eta) = +\infty$, and there is nothing to prove.

Suppose that $\lambda \ll \eta$, write $\varphi := d\lambda/d\eta$, and define

$$\tilde{\lambda}(du) := \varphi(T(u)) \mu(du).$$

Then

$$\tilde{\lambda} \circ T^{-1} = \lambda, \quad \text{Ent}(\tilde{\lambda} \mid \mu) = \text{Ent}(\lambda \mid \eta).$$

Moreover, $\tilde{\lambda}(K) = \mu(K) = 1$. Hence every coupling π of $\tilde{\lambda}$ and μ is concentrated on $K \times K$, and therefore

$$\int |T(u) - T(v)|^p \pi(du, dv) \leq L_K(T)^p \int |u - v|^p \pi(du, dv).$$

Taking the infimum over π gives

$$W_p(\lambda, \eta) \leq L_K(T) W_p(\tilde{\lambda}, \mu).$$

Consequently,

$$\begin{aligned} W_p(\lambda, \eta)^2 &\leq L_K(T)^2 W_p(\tilde{\lambda}, \mu)^2 \\ &\leq 2C_T L_K(T)^2 \text{Ent}(\tilde{\lambda} \mid \mu) \\ &= 2C_T L_K(T)^2 \text{Ent}(\lambda \mid \eta). \end{aligned}$$

□

2.3. Proof of the compact-set Jacobian estimate. The proof of Theorem 2.2 relies on two elementary estimates for linear stochastic equations.

Lemma 2.1. *Let Y_t solve*

$$\begin{cases} dY_t = A_t Y_t dt + \sum_{\ell=1}^r C_{\ell,t} Y_t dB_t^\ell, \\ Y_0 = y. \end{cases}$$

Assume that, for every $z \in \mathbb{R}^d$,

$$\langle z, A_t z \rangle \leq L_x |z|^2, \quad \sum_{\ell=1}^r |C_{\ell,t} z|^2 \leq L_\sigma^2 |z|^2.$$

Then, for every $q \geq 1$,

$$\mathbb{E}|Y_t|^{2q} \leq |y|^{2q} \exp \{q(2L_x + (2q-1)L_\sigma^2)t\}.$$

Consequently,

$$\mathbb{E}\|J_t(u)\|_{\text{op}}^{2q} \leq d^q \exp \{q(2L_x + (2q-1)L_\sigma^2)t\}.$$

Proof. Applying Itô's formula to $|Y_t|^{2q}$, after a standard localization, gives

$$\begin{aligned} d|Y_t|^{2q} &= 2q|Y_t|^{2q-2} \langle Y_t, A_t Y_t \rangle dt + q|Y_t|^{2q-2} \sum_{\ell=1}^r |C_{\ell,t} Y_t|^2 dt \\ &\quad + 2q(q-1)|Y_t|^{2q-4} \sum_{\ell=1}^r \langle Y_t, C_{\ell,t} Y_t \rangle^2 dt + dM_t, \end{aligned}$$

where M_t is a local martingale. Since

$$\langle Y_t, C_{\ell,t} Y_t \rangle^2 \leq |Y_t|^2 |C_{\ell,t} Y_t|^2,$$

the assumptions imply

$$d|Y_t|^{2q} \leq q(2L_x + (2q-1)L_\sigma^2)|Y_t|^{2q} dt + dM_t.$$

Taking expectations in the localized equation and applying Gronwall's lemma yields

$$\mathbb{E}|Y_t|^{2q} \leq |y|^{2q} \exp \{q(2L_x + (2q-1)L_\sigma^2)t\}.$$

For the matrix estimate, apply the vector estimate to $J_t(u)e_j$, $j = 1, \dots, d$, and use

$$\|J_t(u)\|_{\text{op}}^{2q} \leq \|J_t(u)\|_{\text{HS}}^{2q} = \left(\sum_{j=1}^d |J_t(u)e_j|^2 \right)^q \leq d^{q-1} \sum_{j=1}^d |J_t(u)e_j|^{2q}.$$

This gives

$$\mathbb{E}\|J_t(u)\|_{\text{op}}^{2q} \leq d^q \exp \{q(2L_x + (2q-1)L_\sigma^2)t\}.$$

□

Lemma 2.2. *Let $m \geq 2$ be an even integer, and suppose that the matrix-valued process D_t satisfies*

$$dD_t = A_t D_t dt + \sum_{\ell=1}^r C_{\ell,t} D_t dB_t^\ell + F_t dt + \sum_{\ell=1}^r G_{\ell,t} dB_t^\ell.$$

Assume that, for every $M \in \mathbb{R}^{d \times d}$,

$$\langle M, A_t M \rangle_{\text{HS}} \leq L_x \|M\|_{\text{HS}}^2, \quad \sum_{\ell=1}^r \|C_{\ell,t} M\|_{\text{HS}}^2 \leq L_\sigma^2 \|M\|_{\text{HS}}^2.$$

Then there exist constants $C_m, c_m < \infty$, depending only on m, L_x, L_σ , such that

$$\begin{aligned} \mathbb{E} \|D_t\|_{\text{HS}}^m &\leq e^{c_m t} \mathbb{E} \|D_0\|_{\text{HS}}^m \\ &\quad + C_m \int_0^t e^{c_m(t-s)} \left[\mathbb{E} \|F_s\|_{\text{HS}}^m + \mathbb{E} \left(\sum_{\ell=1}^r \|G_{\ell,s}\|_{\text{HS}}^2 \right)^{m/2} \right] ds. \end{aligned}$$

Proof of Theorem 2.2. Fix a compact set $K \subset \mathbb{R}^d$, $t \geq 0$, and $p \geq 1$. Choose an even integer $m > \max\{2p, 2d\}$.

Fix $u, v \in \mathbb{R}^d$ and set

$$\Delta X_t := X_t(u) - X_t(v).$$

Since the same random measure μ_t appears in both equations, subtraction yields

$$\begin{aligned} d\Delta X_t &= (a(X_t(u), \mu_t) - a(X_t(v), \mu_t)) dt \\ &\quad + \sum_{\ell=1}^r (\sigma_\ell(X_t(u)) - \sigma_\ell(X_t(v))) dB_t^\ell. \end{aligned}$$

Define

$$\tilde{A}_t(u, v) := \int_0^1 \partial_x a(X_t(v) + \theta \Delta X_t, \mu_t) d\theta$$

and, for $\ell = 1, \dots, r$,

$$\tilde{C}_{\ell,t}(u, v) := \int_0^1 \partial_x \sigma_\ell(X_t(v) + \theta \Delta X_t) d\theta.$$

By the fundamental theorem of calculus,

$$\begin{cases} d\Delta X_t = \tilde{A}_t(u, v) \Delta X_t dt + \sum_{\ell=1}^r \tilde{C}_{\ell,t}(u, v) \Delta X_t dB_t^\ell, \\ \Delta X_0 = u - v. \end{cases}$$

Moreover, by (4),

$$\|\tilde{A}_t(u, v)\|_{\text{op}} \leq L_x.$$

Jensen's inequality also gives, for every $z \in \mathbb{R}^d$,

$$\sum_{\ell=1}^r |\tilde{C}_{\ell,t}(u, v) z|^2 \leq \int_0^1 \sum_{\ell=1}^r |\partial_x \sigma_\ell(X_t(v) + \theta \Delta X_t) z|^2 d\theta \leq L_\sigma^2 |z|^2.$$

Applying Lemma 2.1 with $q = m/2$ and with $q = m$, respectively, yields

$$(12) \quad \mathbb{E} |X_t(u) - X_t(v)|^m \leq C_m e^{c_m t} |u - v|^m,$$

$$(13) \quad \mathbb{E} |X_t(u) - X_t(v)|^{2m} \leq C_m e^{c_m t} |u - v|^{2m}.$$

Lemma 2.1 gives, for every $q \geq 1$ and every $u \in \mathbb{R}^d$,

$$\mathbb{E} \|J_t(u)\|_{\text{op}}^{2q} \leq d^q \exp \left(q(2L + dL_\sigma^2)t + 2q^2 dL_\sigma^2 t \right).$$

In particular, taking $q = m/2$ and $q = m$, we obtain

$$(14) \quad \mathbb{E} \|J_t(u)\|_{\text{op}}^m \leq C_m e^{c_m t}, \quad \mathbb{E} \|J_t(u)\|_{\text{op}}^{2m} \leq C_m e^{c_m t},$$

uniformly in $u \in \mathbb{R}^d$.

Fix $u, v \in \mathbb{R}^d$, and write

$$D_t := J_t(u) - J_t(v).$$

Set

$$A_t(w) := \partial_x a(X_t(w), \mu_t), \quad C_{\ell,t}(w) := \partial_x \sigma_\ell(X_t(w)), \quad \ell = 1, \dots, r.$$

Subtracting the Jacobian equations at u and v , we obtain

$$(15) \quad \begin{cases} dD_t = A_t(u)D_t dt + \sum_{\ell=1}^r C_{\ell,t}(u)D_t dB_t^\ell + F_t dt + \sum_{\ell=1}^r G_{\ell,t} dB_t^\ell, \\ D_0 = 0, \end{cases}$$

where

$$F_t := (A_t(u) - A_t(v))J_t(v)$$

and

$$G_{\ell,t} := (C_{\ell,t}(u) - C_{\ell,t}(v))J_t(v), \quad \ell = 1, \dots, r.$$

For every $M \in \mathbb{R}^{d \times d}$,

$$\langle M, A_t(u)M \rangle_{\text{HS}} \leq L_x \|M\|_{\text{HS}}^2,$$

and

$$\sum_{\ell=1}^r \|C_{\ell,t}(u)M\|_{\text{HS}}^2 \leq L_\sigma^2 \|M\|_{\text{HS}}^2.$$

Indeed, both inequalities follow by applying (4) to the columns of M .

Similarly, applying (6) columnwise gives

$$(16) \quad \|F_t\|_{\text{HS}} \leq H |\Delta X_t| \|J_t(v)\|_{\text{HS}}$$

and

$$\left(\sum_{\ell=1}^r \|G_{\ell,t}\|_{\text{HS}}^2 \right)^{1/2} \leq H |\Delta X_t| \|J_t(v)\|_{\text{HS}}.$$

Consequently, Hölder's inequality, (13), and (14) imply

$$\begin{aligned} \mathbb{E} \|F_t\|_{\text{HS}}^m + \mathbb{E} \left(\sum_{\ell=1}^r \|G_{\ell,t}\|_{\text{HS}}^2 \right)^{m/2} \\ \leq C_m (\mathbb{E} |\Delta X_t|^{2m})^{1/2} (\mathbb{E} \|J_t(v)\|_{\text{HS}}^{2m})^{1/2} \\ \leq C_m e^{c_m t} |u - v|^m. \end{aligned}$$

Here we also used $\|M\|_{\text{HS}} \leq \sqrt{d} \|M\|_{\text{op}}$.

Applying Lemma 2.2 to (15), and using $D_0 = 0$, gives

$$\begin{aligned} \mathbb{E} \|D_t\|_{\text{HS}}^m &\leq C_m |u - v|^m \int_0^t e^{c_m(t-s)} e^{c_m s} ds \\ &\leq C_m e^{c_m t} |u - v|^m, \end{aligned}$$

after enlarging c_m . Therefore,

$$(17) \quad \mathbb{E} \|J_t(u) - J_t(v)\|_{\text{HS}}^m \leq C_m e^{c_m t} |u - v|^m, \quad u, v \in \mathbb{R}^d.$$

Let $Q \subset \mathbb{R}^d$ be a compact cube containing K , and choose $u_0 \in K$. By (17), $u, v \in Q$

$$\mathbb{E} \|J_t(u) - J_t(v)\|_{\text{HS}}^m \leq C_m e^{c_m t} |u - v|^m.$$

Since $m > d$, this may be written as

$$\mathbb{E} \|J_t(u) - J_t(v)\|_{\text{HS}}^m \leq C_m e^{c_m t} |u - v|^{d+(m-d)}.$$

Hence the quantitative Kolmogorov–Totoki theorem, applied to the $\mathbb{R}^{d \times d}$ -valued random field

$$u \mapsto J_t(u) - J_t(u_0),$$

gives

$$(18) \quad \mathbb{E} \left[\sup_{u \in Q} \|J_t(u) - J_t(u_0)\|_{\text{HS}}^m \right] \leq C_{m,Q} e^{c_m t}.$$

Here we identify $\mathbb{R}^{d \times d}$ with $\mathbb{R}^{d^2}$.

For every $u \in K$,

$$\|J_t(u)\|_{\text{op}} \leq \|J_t(u_0)\|_{\text{op}} + \|J_t(u) - J_t(u_0)\|_{\text{HS}}.$$

Consequently,

$$\mathcal{K}_t(K) \leq \|J_t(u_0)\|_{\text{op}} + \sup_{u \in Q} \|J_t(u) - J_t(u_0)\|_{\text{HS}}.$$

Using (14), (18), and $(a + b)^m \leq 2^{m-1}(a^m + b^m)$, we obtain

$$\mathbb{E}\mathcal{K}_t(K)^m \leq C_{m,K}e^{c_m t}.$$

Finally,

$$\mathbb{E}\mathcal{K}_t(K)^p \leq (\mathbb{E}\mathcal{K}_t(K)^m)^{p/m} \leq C_{p,K}e^{c_p t}.$$

This proves (7). □

3. THE AVERAGED JACOBIAN

The preceding pathwise estimates use the quantity $\mathcal{K}_t(K)$ which controls the worst possible stretching of the flow on the compact set K . Consider instead the averaged Jacobian for the transported measure $\mu_t = \mu_0 \circ T_t^{-1}$,

$$\mathcal{E}_t := \int_{\mathbb{R}^d} \|J_t(u)\|_{\text{HS}}^2 \mu_0(du).$$

It measures average deformation of the flow over the initial mass, rather than the maximal deformation on the whole support.

Proposition 3.1 (Averaged Jacobian energy estimate). *Assume that the flow is differentiable in the initial label and that there exist constants $L_{\text{sym}}, \Gamma < \infty$ such that*

$$(19) \quad \langle z, \partial_x a(x, \mu) z \rangle \leq L_{\text{sym}} |z|^2, \quad x, z \in \mathbb{R}^d, \quad \mu \in \mathcal{P}_2(\mathbb{R}^d),$$

and

$$(20) \quad \sum_{\ell=1}^r |\partial_x \sigma_\ell(x) z|^2 \leq \Gamma^2 |z|^2, \quad x, z \in \mathbb{R}^d.$$

Then

$$(21) \quad \mathbb{E} \mathcal{E}_t \leq d \exp\{(2L_{\text{sym}} + \Gamma^2)t\}.$$

Proof. Recall that

$$dJ_t(u) = A_t(u) J_t(u) dt + \sum_{\ell=1}^r C_{\ell,t}(u) J_t(u) dB_t^\ell,$$

where

$$A_t(u) = \partial_x a(X_t(u), \mu_t), \quad C_{\ell,t}(u) = \partial_x \sigma_\ell(X_t(u)).$$

Applying Itô's formula to $\|J_t(u)\|_{\text{HS}}^2$ gives

$$\begin{aligned} d\|J_t(u)\|_{\text{HS}}^2 &= \left[2\langle J_t(u), A_t(u) J_t(u) \rangle_{\text{HS}} + \sum_{\ell=1}^r \|C_{\ell,t}(u) J_t(u)\|_{\text{HS}}^2 \right] dt \\ &\quad + 2 \sum_{\ell=1}^r \langle J_t(u), C_{\ell,t}(u) J_t(u) \rangle_{\text{HS}} dB_t^\ell. \end{aligned}$$

Since

$$\langle J, A_t J \rangle_{\text{HS}} = \sum_{i=1}^d \langle J e_i, A_t J e_i \rangle,$$

condition equation (19) gives

$$2\langle J, A_t J \rangle_{\text{HS}} \leq 2L_{\text{sym}} \|J\|_{\text{HS}}^2.$$

Similarly, equation (20) gives

$$\sum_{\ell=1}^r \|C_{\ell,t} J\|_{\text{HS}}^2 = \sum_{i=1}^d \sum_{\ell=1}^r |C_{\ell,t} J e_i|^2 \leq \Gamma^2 \|J\|_{\text{HS}}^2.$$

Integrating with respect to $\mu_0(du)$ and taking expectations removes the martingale term, hence

$$\frac{d}{dt} \mathbb{E} \mathcal{E}_t \leq (2L_{\text{sym}} + \Gamma^2) \mathbb{E} \mathcal{E}_t.$$

Since $J_0(u) = I$, we have $\mathcal{E}_0 = \|I\|_{\text{HS}}^2 = d$. Gronwall's lemma proves equation (21). $\square$

To estimate $\mathbb{E} \sup_{u \in K} \|J_t(u)\|^p$, one must control increments $J_t(u) - J_t(v)$, which requires assumptions on the spatial regularity of $\partial_x a$ and $\partial_x \sigma$. The averaged quantity $\mathcal{E}_t$ does not compare two labels. It only estimates each $J_t(u)$ and integrates in u , so the one-sided bounds equations (19) and (20) are sufficient.

4. INTERNAL FLUCTUATIONS

This section concerns a separate kernel-type interacting model. In contrast to (1), both the drift and diffusion may depend on the order parameter Θ_t . Consequently, the assumptions of sections 2 and 3 are not imposed here.

$$(22) \quad \begin{cases} dX_t(u) &= \left[b(X_t(u), \Theta_t) + \int_{\mathbb{R}^d} F(X_t(u), X_t(v), \Theta_t) \mu_0(dv) \right] dt \\ &\quad + \sum_{\ell=1}^r \sigma_\ell(X_t(u), \Theta_t) dB_t^\ell, \\ X_0(u) &= u, \quad u \in \mathbb{R}^d, \\ \mu_t &= \mu_0 \circ X_t^{-1}. \end{cases}$$

Let $q \geq 1$, and let

$$\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^q, \quad b : \mathbb{R}^d \times \mathbb{R}^q \rightarrow \mathbb{R}^d, \quad F : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^q \rightarrow \mathbb{R}^d,$$

with $\sigma_\ell : \mathbb{R}^d \times \mathbb{R}^q \rightarrow \mathbb{R}^d$, $\ell = 1, \dots, r$. Define

$$\theta(\mu) := \int_{\mathbb{R}^d} \Phi(x) \mu(dx), \quad \Theta_t := \theta(\mu_t) = \int_{\mathbb{R}^d} \Phi(X_t(v)) \mu_0(dv).$$

Assumption 3 (Well-posedness of Equation (22)). *The maps b, F, σ_ℓ, Φ are globally Lipschitz and have at most linear growth. More explicitly, there exists $L < \infty$ such that*

$$|\Phi(x) - \Phi(y)| \leq L|x - y|,$$

$$|b(x, \theta) - b(y, \eta)| \leq L(|x - y| + |\theta - \eta|),$$

$$|F(x, z, \theta) - F(y, w, \eta)| \leq L(|x - y| + |z - w| + |\theta - \eta|),$$

and

$$\sum_{\ell=1}^r |\sigma_\ell(x, \theta) - \sigma_\ell(y, \eta)|^2 \leq L^2(|x - y|^2 + |\theta - \eta|^2).$$

Under Assumption 3, equation (22) admits a unique global continuous solution.

This section studies the pairwise energy for (22)

$$\mathcal{D}_t := \iint_{\mathbb{R}^d \times \mathbb{R}^d} |X_t(u) - X_t(v)|^2 \mu_0(du) \mu_0(dv).$$

4.1. Dissipativity assumptions. Equivalently, if

$$\bar{X}_t := \int_{\mathbb{R}^d} X_t(u) \mu_0(du),$$

and

$$\mathcal{V}_t := \int_{\mathbb{R}^d} |X_t(u) - \bar{X}_t|^2 \mu_0(du),$$

then

$$\mathcal{D}_t = 2\mathcal{V}_t.$$

Indeed,

$$\begin{aligned} \mathcal{D}_t &= \iint |X_t(u) - X_t(v)|^2 \mu_0(du) \mu_0(dv) \\ &= \iint (|X_t(u)|^2 + |X_t(v)|^2 - 2\langle X_t(u), X_t(v) \rangle) \mu_0(du) \mu_0(dv) \\ &= 2 \int |X_t(u)|^2 \mu_0(du) - 2 \left| \int X_t(u) \mu_0(du) \right|^2 \\ &= 2 \int |X_t(u) - \bar{X}_t|^2 \mu_0(du) = 2\mathcal{V}_t. \end{aligned}$$

We assume the following dissipativity condition.

Assumption 4 (Pairwise dissipativity). *There exists $\lambda > 0$ such that for every $x, y \in \mathbb{R}^d$, every $\theta \in \mathbb{R}^q$, and every $\nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$2 \left\langle x - y, b(x, \theta) - b(y, \theta) + \int_{\mathbb{R}^d} [F(x, z, \theta) - F(y, z, \theta)] \nu(dz) \right\rangle + \sum_{\ell=1}^r |\sigma_\ell(x, \theta) - \sigma_\ell(y, \theta)|^2 \leq -\lambda |x - y|^2.$$

The dissipativity condition covers several basic models.

Example 4.1 (Ornstein–Uhlenbeck SDEWI). Consider

$$dX_t(u) = -\lambda_0(X_t(u) - m_t) dt + \sigma(X_t(u)) dB_t, \quad m_t = \int_{\mathbb{R}^d} X_t(v) \mu_0(dv).$$

If

$$\|\sigma(x) - \sigma(y)\|_{\text{HS}} \leq L_\sigma |x - y|,$$

then

$$\mathbb{E} \mathcal{D}_t \leq e^{-(2\lambda_0 - L_\sigma^2)t} \mathcal{D}_0.$$

Thus the cloud contracts whenever $2\lambda_0 > L_\sigma^2$.

Example 4.2 (Gradient systems). Take

$$b(x, \theta) = -\partial V(x), \quad F(x, y, \theta) = -\partial W(x - y).$$

Then

$$dX_t(u) = \left[-\partial V(X_t(u)) - \int_{\mathbb{R}^d} \partial W(X_t(u) - X_t(v)) \mu_0(dv) \right] dt + \sigma(X_t(u)) dB_t.$$

If

$$\partial_x^2 V \geq \rho I, \quad \partial_x^2 W \geq \rho_W I, \quad \|\sigma(x) - \sigma(y)\| \leq L_\sigma |x - y|,$$

then

$$\mathbb{E} \mathcal{D}_t \leq e^{-(2\rho + 2\rho_W - L_\sigma^2)t} \mathcal{D}_0.$$

Example 4.3 (Nonlinear statistics). Let

$$\Theta_t = \int_{\mathbb{R}^d} \psi(X_t(v)) \mu_0(dv),$$

and consider

$$dX_t(u) = -\lambda_0(X_t(u) - \Theta_t) dt + \sigma(X_t(u), \Theta_t) dB_t.$$

For two labels u, v , the order parameter cancels in the drift difference:

$$-\lambda_0(X_t(u) - \Theta_t) + \lambda_0(X_t(v) - \Theta_t) = -\lambda_0(X_t(u) - X_t(v)).$$

4.2. Proof.

Theorem 4.1. *Assume that $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$ and that Assumption 4 holds. Then*

$$\mathbb{E} \mathcal{D}_t \leq e^{-\lambda t} \mathcal{D}_0.$$

In particular,

$$\mathbb{E} W_2^2(\mu_t, \delta_{\bar{X}_t}) \leq e^{-\lambda t} W_2^2(\mu_0, \delta_{\bar{X}_0}).$$

Thus the transported measure μ_t concentrates around its random center of mass $\bar{X}_t$.

Proof. Fix two labels $u, v \in \mathbb{R}^d$. Define the difference

$$Z_t^{u,v} := X_t(u) - X_t(v).$$

We first prove that under Assumption 4

$$\mathbb{E} |Z_t^{u,v}|^2 \leq e^{-\lambda t} |u - v|^2,$$

then we integrate this estimate with respect to $\mu_0(du)\mu_0(dv)$.

Writing equation (22) for $X_t(u)$ and for $X_t(v)$ and then subtracting the two equations gives

$$dZ_t^{u,v} = G_t^{u,v} dt + \sum_{\ell=1}^r H_{\ell,t}^{u,v} dB_t^\ell,$$

where the drift coefficient

$$\begin{aligned} G_t^{u,v} &:= b(X_t(u), \Theta_t) - b(X_t(v), \Theta_t) \\ &+ \int_{\mathbb{R}^d} [F(X_t(u), X_t(w), \Theta_t) - F(X_t(v), X_t(w), \Theta_t)] \mu_0(dw), \end{aligned}$$

and the diffusion coefficient is

$$H_{\ell,t}^{u,v} := \sigma_\ell(X_t(u), \Theta_t) - \sigma_\ell(X_t(v), \Theta_t).$$

Since $\mu_t = \mu_0 \circ X_t^{-1}$, we may rewrite the interaction term as

$$\int_{\mathbb{R}^d} F(X_t(u), X_t(w), \Theta_t) \mu_0(dw) = \int_{\mathbb{R}^d} F(X_t(u), z, \Theta_t) \mu_t(dz).$$

Thus

$$\begin{aligned} G_t^{u,v} &:= b(X_t(u), \Theta_t) - b(X_t(v), \Theta_t) \\ &+ \int_{\mathbb{R}^d} [F(X_t(u), z, \Theta_t) - F(X_t(v), z, \Theta_t)] \mu_t(dz). \end{aligned}$$

Applying Itô's formula to $|Z_t^{u,v}|^2$, we obtain

$$\begin{aligned} d|Z_t^{u,v}|^2 &= 2\langle Z_t^{u,v}, G_t^{u,v} \rangle dt + \sum_{\ell=1}^r |H_{\ell,t}^{u,v}|^2 dt \\ &+ 2 \sum_{\ell=1}^r \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell. \end{aligned}$$

Apply Assumption 4 with $x = X_t(u)$, $y = X_t(v)$, $\theta = \Theta_t$, and $\nu = \mu_t$. Then

$$2\langle Z_t^{u,v}, G_t^{u,v} \rangle + \sum_{\ell=1}^r |H_{\ell,t}^{u,v}|^2 \leq -\lambda |Z_t^{u,v}|^2.$$

Therefore

$$(23) \quad d|Z_t^{u,v}|^2 \leq -\lambda |Z_t^{u,v}|^2 dt + 2 \sum_{\ell=1}^r \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell.$$

Set the new variable

$$Y_t^{u,v} := e^{\lambda t} |Z_t^{u,v}|^2.$$

By Itô's formula,

$$\begin{aligned} dY_t^{u,v} &= e^{\lambda t} \left[\lambda |Z_t^{u,v}|^2 + 2\langle Z_t^{u,v}, G_t^{u,v} \rangle \right. \\ &\quad \left. + \sum_{\ell=1}^r |H_{\ell,t}^{u,v}|^2 \right] dt \\ &\quad + 2e^{\lambda t} \sum_{\ell=1}^r \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell. \end{aligned}$$

By the drift bound equation (23), the drift term above is nonpositive. Hence

$$dY_t^{u,v} \leq 2e^{\lambda t} \sum_{\ell=1}^r \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell.$$

To make this completely rigorous, introduce the localization sequence

$$\tau_R := \inf\{t \geq 0 : |X_t(u)| + |X_t(v)| + |\Theta_t| \geq R\}.$$

Since the solution is global and continuous, $\tau_R \uparrow \infty$ almost surely as $R \rightarrow \infty$. Stopping this inequality at τ_R , we get

$$Y_{t \wedge \tau_R}^{u,v} \leq Y_0^{u,v} + 2 \sum_{\ell=1}^r \int_0^{t \wedge \tau_R} e^{\lambda s} \langle Z_s^{u,v}, H_{\ell,s}^{u,v} \rangle dB_s^\ell.$$

The stopped stochastic integral is a true martingale. Therefore, taking expectations gives

$$\mathbb{E} Y_{t \wedge \tau_R}^{u,v} \leq Y_0^{u,v}.$$

Since $Y_0^{u,v} = |u - v|^2$, we have

$$\mathbb{E} \left[e^{\lambda(t \wedge \tau_R)} |Z_{t \wedge \tau_R}^{u,v}|^2 \right] \leq |u - v|^2.$$

Letting $R \rightarrow \infty$ and using Fatou's lemma yields

$$\mathbb{E} [e^{\lambda t} |Z_t^{u,v}|^2] \leq |u - v|^2.$$

Therefore

$$\mathbb{E} |X_t(u) - X_t(v)|^2 \leq e^{-\lambda t} |u - v|^2.$$

This proves the pairwise synchronization estimate.

Now integrate the pairwise synchronization estimate with respect to $\mu_0(du)\mu_0(dv)$. Since $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{D}_0 = \iint |u - v|^2 \mu_0(du)\mu_0(dv) < \infty.$$

By Fubini's theorem,

$$\begin{aligned} \mathbb{E} \mathcal{D}_t &= \mathbb{E} \iint |X_t(u) - X_t(v)|^2 \mu_0(du)\mu_0(dv) \\ &= \iint \mathbb{E} |X_t(u) - X_t(v)|^2 \mu_0(du)\mu_0(dv) \\ &\leq e^{-\lambda t} \iint |u - v|^2 \mu_0(du)\mu_0(dv). \end{aligned}$$

Hence $\mathbb{E} \mathcal{D}_t \leq e^{-\lambda t} \mathcal{D}_0$, proving the first estimate in the theorem.

Therefore

$$\mathbb{E} \mathcal{V}_t = \frac{1}{2} \mathbb{E} \mathcal{D}_t \leq \frac{1}{2} e^{-\lambda t} \mathcal{D}_0 = e^{-\lambda t} \mathcal{V}_0.$$

This proves the internal fluctuation decay estimate.

Finally, since $\mu_t = \mu_0 \circ X_t^{-1}$,

$$\int_{\mathbb{R}^d} |x - \bar{X}_t|^2 \mu_t(dx) = \int_{\mathbb{R}^d} |X_t(u) - \bar{X}_t|^2 \mu_0(du) = \mathcal{V}_t.$$

For the Dirac measure $\delta_{\bar{X}_t}$,

$$W_2^2(\mu_t, \delta_{\bar{X}_t}) = \int_{\mathbb{R}^d} |x - \bar{X}_t|^2 \mu_t(dx).$$

Therefore

$$W_2^2(\mu_t, \delta_{\bar{X}_t}) = \mathcal{V}_t.$$

Taking expectations and using the previous estimate gives

$$\mathbb{E} W_2^2(\mu_t, \delta_{\bar{X}_t}) \leq e^{-\lambda t} W_2^2(\mu_0, \delta_{\bar{X}_0}).$$

□

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